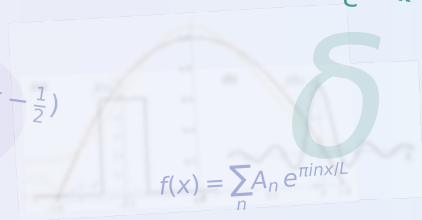


$$e^{2\pi i \mathbf{k} \cdot \mathbf{r}}$$

$$\int$$

$$\Pi(x) = H(x + \frac{1}{2}) - H(x - \frac{1}{2})$$

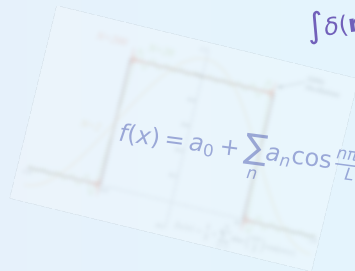


$$e^{2\pi i \mathbf{k} \cdot \mathbf{r}}$$

$$\delta$$

$$f(x) = \sum_n A_n e^{\pi i n x / L}$$

$$\int \delta(\mathbf{r} - \mathbf{a}) d\mathbf{r} = 1$$



$$f(x) = a_0 + \sum_n a_n \cos \frac{n\pi x}{L} + \sum_n b_n \sin \frac{n\pi x}{L}$$

$$\Pi(x) = H(x + \frac{1}{2}) - H(x - \frac{1}{2})$$

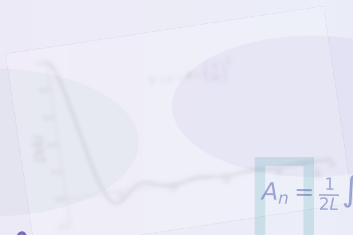
$$A_n = \frac{1}{2L} \int_{-L}^{+L} f(x) e^{-\pi i n x / L} dx$$

$$\sum$$

$$f(\mathbf{r}) \leftrightarrow F(\mathbf{k})$$

$$f(x) = \sum_n A_n e^{\pi i n x / L}$$

$$f(\mathbf{r}) \leftrightarrow F(\mathbf{k})$$



$$\Pi$$

$$A_n = \frac{1}{2L} \int_{-L}^{+L} f(x) e^{-\pi i n x / L} dx$$

$$\delta(x) = \lim_{a \rightarrow \infty} \frac{a}{\sqrt{\pi}} e^{-a^2 x^2}$$

$$f(x) = a_0 + \sum_n a_n \cos \frac{n\pi x}{L} + \sum_n b_n \sin \frac{n\pi x}{L}$$

$$\int \delta(\mathbf{r} - \mathbf{a}) d\mathbf{r} = 1$$

$$\delta(x) = \lim_{a \rightarrow \infty} \frac{a}{\sqrt{\pi}} e^{-a^2 x^2}$$